

Preserving Virtual Reality

Zeynep Abes

School of Cinematic Arts, University of Southern California, United States of America
zeynep.abes@gmail.com

ABSTRACT

According to the Library of Congress, 75% of silent movies before the 1930s are lost forever. [Pierce, 2013] During the early 20th century, films were not considered to have much future value or lasting significance which lead to the majority of them not being properly archived or preserved. Most films were intentionally destroyed to avoid spending money on storage space and expensive upkeep of the materials. [Pierce, 2013] By the time studios and institutions started to establish film archives, it was already too late for most of early cinema. We are seeing a similar scenario take place with preserving interactive media projects as the majority of VR experiences have been ignored by institutional archives. Preserving virtual reality has a particularly unique challenge because it is designed with interface, hardware, and software in mind. Unlike films and games that can easily be archived in formats that can be universally played on screens, the majority of virtual reality experiences are designed to run on very specific hardware. Once a headset is no longer supported, the dedicated VR experiences also disappear with it, unless they are continuously updated to move to newer systems. Unfortunately, the majority of independent developers and creators do not have the resources, time, or budget to keep up with the constant turnover rate of new hardware. Not to mention the lack of infrastructure that prevents content creators from ensuring the longevity of titles that are not positioned with a financial gain.

The panel will explore and discuss the challenges of establishing a standard of best practices for VR preservation and how to support the futurity of these studies. The discussion will further delve into the need for collaboration between artists, institutions, and above all the technology companies that the medium so heavily relies on. Panelists will provide real-world examples of why it's important to consider the potential long-term value of archiving virtual reality content, both for researchers and for the general public. The panel will identify key challenges and risks to bring clarity to the complex nature of this undertaking, like the intricate ecosystem of hardware and software and how the rapid obsolescence cycles continuously challenge efforts for conservation. We know that archives are not neutral. They are a product of their culture, oftentimes the dominant culture. In this case, big tech is the gatekeeper when it comes to deciding what gets to be kept and what gets lost within the virtual reality community. As companies like Meta take further control of the industry, people will surely not get a clear understanding of the systems of oppression within the history of immersive media, let

alone a virtual reality archive for future students to reference from the past.

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1 PANELISTS

Scott Fisher is Professor of Media Arts + Practice in the USC School of Cinematic Arts. He is the Founding Chair of the Interactive Media Division, former Dean of Research, and Director of the Mobile & Environmental Media Lab. Fisher has worked since the early 80s to develop virtual reality and augmented reality systems at NASA's Ames Research Center, Atari Research Labs, MIT's Architecture Machine Group and at Keio University in Japan. As a Research Scientist at NASA Ames, his lab played a pioneering role in developing several virtual reality technologies that form the basis for products commercially available today.

Henry Lowood is the Harold C. Hohbach Curator at Stanford University, responsible for history of science & technology collections and film & media collections in the Stanford Libraries. He has combined interests in history, technological innovation and the history of digital games and simulations and leads the Silicon Valley Archives at Stanford. He headed Stanford's work on game and virtual world preservation in the multi-institutional Preserving Virtual Worlds project. He co-edits the book series, Game Histories, for MIT Press, and the journal ROMchip. His *Replayed: Essential Writings on Software Preservation and Game Histories* (Johns Hopkins UP) is forthcoming.

Samantha Gorman is an interdisciplinary scholar, artist and educator. She specializes in writing (and narrative design) for digital media, directing for live digital performance/theater, and the theory and practice of immersive media (AR/VR). She is the co-lead of studio Tender Claws and a professor of games in the Art + Design Department in CAMD at Northeastern University.

Erkki Huhtamo works as a professor at the University of California Los Angeles (UCLA), Departments of Design Media Arts, and Film, Television, and Digital Media. He received his Ph.D. in cultural history from the University of Turku, Finland. Huhtamo is an internationally renowned media historian and theorist, and also a specialist in the history and aesthetics of media arts. He is one of the founders of an emerging approach to media studies known as media archaeology.

Ruthie Doyle is an artist and filmmaker based in Los Angeles. Since 2013 she has worked with the Sundance Institute's New Frontier program team, which supports and exhibits artists working at the convergence of film, art, media and technology at the Sundance Film Festival, as well as year-round via Labs and residencies.

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Moderator: Zeynep is an artist, curator and educator from Istanbul, Turkey. She studied film and interactive media at Emerson College and worked in notable institutions like LACMA's Art+Tech lab, and Sundance Film Festival's New Frontier Exhibitions. She is a recent graduate of UCLA's Design Media Arts MFA program and a current Ph.D. candidate in the Media Arts and Practice program at USC's School of Cinematic Arts. She primarily works with archived media, photogrammetry and immersive media. She's in pursuit of

exploring the role artists play in preserving memories to navigate the struggle that arises from changing social environments and shifting identities.

REFERENCES

David Pierce. "The Survival of American Silent Feature Films: 1912–1929." Council on Library & Information Resources. The Library of Congress, September 2013. <https://www.loc.gov/programs/national-film-preservation-board/preservation-research/silent-film-database/>